

Automatic Conversation Transcription and Summarisation using Natural Language Processing

Progress Report

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Abstract

Businesses and organisations across the globe are continuously implementing solutions that utilise ASR (Automatic Speech Recognition) and NLP (Natural Language Processing) to streamline their operations. The solution proposed in this work aims to provide automatic and seamless transcription of audio obtained during meetings, segmenting between two or more speakers within these conversations. Subsequently, meaningful information will be extracted from the resulting transcript that industry users may analyse and utilise to their advantage.

1 Introduction

1.1 Problem Definition

Meetings are an essential component to everyday professional life and are crucial for the creation and sharing of information [45]. Following the COVID-19 pandemic, there has been a shift towards online meetings that are held via Zoom and various other video conference platforms. This shift to remote work, along with advances in speech recognition technology, has led to an increase in the amount of dialogue that is or can be transcribed [50]. The documentation of important conversations, whether they occur virtually or in-person, may prove to be valuable to organisations, with a wide range of applications including updating absentees with new developments, employee training, legal purposes and transparency [6]. In this research, an end-to-end automatic speech recognition system will be proposed, that segments and identifies speakers, returning a transcript and performing a summary of the meeting.

1.2 Motivation

Manual transcription is cumbersome and consuming raw meeting data in its original unstructured format is time consuming and inefficient. Such a system would result in increased productivity and improved accessibility [44]. Applications for conversational speech recognition extend far beyond this domain, including the healthcare sector [9], and such a system could easily be adapted for these use cases. Although there have been significant progress made in individual areas, namely ASR and text summarisation, there has been a limited focus on applying these advanced techniques together to real-world applications [35].

1.3 Aims & Objectives

The overall goal of this project is to deliver a system that contributes to the task of conversation transcription and summarisation, gaining insight into how advanced NLP techniques can be deployed in industry and other real-world applications, while also serving as a solid foundation for future work in this area. This will be achieved via the following objectives:

- Evaluating different approaches and state-of-the-art models in the fields of ASR and Speaker Diarisation in order to build a system that transcribes text with near real-time performance and high accuracy, simultaneously segmenting and identifying between two or more speakers.
- Experimenting with and comparing diverse techniques that are used in academia and industry to summarise text and extract meaningful information from conversation transcripts, ranking conciseness and accuracy.
- Testing the system on corpora related to business meetings and long conversations to evaluate its accuracy, efficiency and weaknesses.

1.4 Why the Problem is Non-Trivial

Difficulty in transcribing and segmenting speech arises due to non-ideal acoustic conditions coupled with regular speech overlaps when dealing with multiple speakers [40]. Overlapped speech has remained virtually unsolved in practical environments for several years, with minimal progress being made [44]. Many studies have attempted to cope with overlapped speech and attempt to separate them into individual channels in order to transcribe them individually [19], however they rely on visual cues obtained from a single or multiple camera array.

Furthermore, the outcome of the summarisation process depends entirely upon the quality of the transcript generated, therefore thorough text enhancement must be carried out. While articles and documents usually possess common linguistic structures, in conversation transcripts the main ideas and arguments are scattered over several speakers and utterances [13]. Therefore, the summarisation of conversational text may prove to be challenging in comparison.

2 Background Research & Literature Review

2.1 Automatic Speech Recognition

Speech is human’s most natural and favoured mode of communication, hence it is no surprise that it has amassed significant interest among researchers as a practical means of input to wide range of machines for a myriad of applications [38]. Automatic speech recognition (ASR), also known as speech-to-text (STT) systems, aims to achieve this goal by converting raw audio waveforms, captured from a microphone as input, to the corresponding text in a given language [28]. This proves to be no straightforward task due to considerable variations among human speech patterns, various accents and dialects as well as diverse factors including gender, speaking style and pacing [14, 28].

Traditionally, Hidden Markov Models (HMMs) were the prevailing speech recognition framework, modelling speech via a stochastic process [15]. This approach generally breaks down the ASR task into independent stages making use of acoustic, pronunciation and language models [33]. In the 1990s and early 2000s, the HMM-Gaussian Mixture Model (GMM) became widely explored for acoustic likelihood evaluation, with HMMs normalising the temporal variability of speech and GMMs calculating the probabilities of HMM states [25]. However, despite extensive research into this approach, the resultant models yielded a performance which was far from desirable for use within practical settings [39]. Hinton et al. argue that this is due to the fact that GMMs are “statistically inefficient for modelling data that lie on or near a nonlinear manifold in the data space” [18].

In the past decade, increased computational power as well as substantial amounts of training data have paved the way for significant developments in the field of ASR arising from the application of deep learning techniques in this area [39]. Deep Neural Networks (DNNs), which are feed-forward artificial neural networks comprising several hidden layers, began to surpass HMM-GMM on several speech recognition benchmarks [18]. Subsequently, the DNN-HMM hybrid architecture emerged as a new superior approach to the large vocabulary continuous speech recognition (LVCSR) problem, while retaining the same individual components as HMM-GMM [11]. Since then, deep learning has been widely researched and applied in industry [39].

More recently, the ASR community has shifted towards end-to-end (E2E) deep learning models, which utilise a single network to convert speech directly from an input sequence to an output token [23]. This single network is more compact and simplified than a complex DNN hybrid model, allowing for deployment to devices with low latency while retaining accuracy. In order to optimise the entire network, a single objective function is required rather than optimising several components individually, thus ensuring that a global optimum is reached. For these reasons, the E2E approach

generally outclasses DNN hybrid models in terms of accuracy in both industry and academia [42].

Numerous E2E techniques exist, the most prevalent of which are Connectionist Temporal Classification (CTC) [41], Recurrent Neural Network Transducer (RNN-T) [16] and Attention-based Encoder-Decoder (AED) [10]. CTC was first proposed by Graves et al. as an E2E ASR system, mapping speech input to output labels automatically with no intermediary processes [17]. CTC first exhibited promising results by Amodei et al. in Deep Speech 2 [2] and Miao et al. in EESSEN [25]. In the latter, acoustic modelling was simplified from a complex HMM-GMM architecture into a single Recurrent Neural Network (RNN) utilising the CTC objective function, hence eliciting the E2E advantages discussed above. For this reason, I decided that it was important to continue to explore how E2E techniques such as CTC have pushed the boundaries of ASR in recent literature.

The current state-of-the-art performance has been achieved by means of semi-supervised learning (SSL) techniques [47], the aim of which is to utilise a large amount of unlabelled data to enhance the performance of a supervised task powered by a small amount of labelled data [48]. This paradigm alleviates the reliance on labelled data, resulting a large quantities of previously unusable data becoming accessible for training [34]. Moreover, it paves the way for the development of robust ASR systems for low-resource languages and dialects which lack adequate annotated datasets, as well as languages such as Japanese and Thai which prove to be problematic for conventional ASR systems due to the lack of spaces separating words. This approach has also contributed to advancements in state-of-the-art in fields such as image classification [7], video processing [30] and others, demonstrating the potential of SSL to benefit downstream applications where data proves to be challenging or impractical to collect. In the field of ASR, where considerable effort is necessary to obtain labelled data, the need arises for a feasible system that tackles this issue [3]. This has been achieved by exploiting large quantities of unlabelled audio data to learn speech representations followed by utilising limited amounts of labelled data to increase performance on a downstream applied task [34].

In particular, Wav2Vec 2.0, introduced by Baevski et al. from Facebook AI, is an E2E model that utilises SSL to break new ground in ASR [3]. Similar to the Masked Language Modelling (MLM) approach used by Bidirectional Encoder Representations from Transformers (BERT) [12], the model is trained by masking fragments of raw audio and predicting speech units for these segments. Since raw audio is a continuous signal comprising various components of sound apart from speech, such as background noise, it is vital that high-level contextualised representations are learnt in order to make Wav2Vec 2.0 more robust and suitable for audio recorded in a multitude of environments. This is achieved by learning basic units of 25ms in length. As a result, Wav2Vec

2.0 outperforms the state-of-the-art even with a fraction of the amount of training data.

Considering the overall shift in E2E models in the field of ASR, it has been decided to go ahead with the SSL E2E approach, utilising the state-of-the-art to maximise efficiency and accuracy in the ASR task. Wav2Vec 2.0 will be selected for this purpose.

2.2 Speaker Diarisation

Speaker diarisation is the task of dividing audio streams containing speech into homogeneous segments corresponding to speech turns and speaker identity. Simply put, it solves the “who spoke when?” problem in a multi-speaker environment [20, 29]. While it has its stand-alone uses, diarisation may be coupled with ASR systems in order to deliver a richer transcript that is more readable and useful for enabling other tasks such as machine translation and summarisation [37]. Its many applications in scenarios such as meeting transcriptions and conversational AI have rendered it an area of substantial interest for researchers and industry users [1]. As is the case for ASR and other areas of speech research and NLP, deep learning has instigated considerable developments in the domain of diarisation [29].

Traditionally, the process of speaker diarisation comprises numerous sub-tasks that together form a pipeline. This conventionally consists of detecting speech activity, segmenting these speech regions into smaller chunks, extracting embeddings from these segments, clustering them based on their correspondence to a respective speaker, and finally re-segmenting them [29]. However, this paradigm elicits two main disadvantages, the first of which is that the pipeline’s modules are optimised separately, with various models often being trained on different datasets with distinct objectives. Secondly, this approach struggles to tackle overlapped speech, since the clustering method used can only assign one speaker label to a particular speech segment [20].

In a similar fashion to ASR, there has been a shift towards E2E speaker diarisation systems as opposed to the conventional modular paradigm [21]. This approach eliminates the inconvenience and complexity of optimising each unit separately, and is also able to tackle overlapped speech. However, these E2E systems are limited to a fixed number of speakers and have only been tested on synthetic short recordings. Moreover, they do not currently push the boundaries of state-of-the-art, which demonstrates that there is much work yet to be done in the world of E2E speaker diarisation.

For these reasons, although utilisation in this field is promising, the E2E paradigm will not be adopted by the system proposed in this research. Instead, a clustering-based pipeline will be deployed. Currently, few open-source toolkits which address this task which each have their strengths and weaknesses [4]. Kaldi [31] contains some diarisation modules but mainly focuses on ASR tasks and is not built using Python,

while S4D [5] focuses on traditional non-deep learning approaches. Therefore, the toolkit chosen to work with is pyannote.audio [4] developed by Bredin et al., which can be used to jointly optimise a single pipeline, and offers pre-trained models achieving state-of-the-art performance for many of the sub-tasks involved, with the possibility to fine-tune.

2.3 Text Summarisation

The task of text summarisation is to successfully condense the most relevant parts of a given text in a concise and comprehensible manner [13]. It is important to distinguish between *abstractive* and *extractive* techniques of summarisation. In *extractive* summarisation, sentences or phrases are ranked or classified in terms of importance and the relevant elements of the given text are subsequently concatenated together to form a summary [26]. Meanwhile, the term *abstractive* denotes a summarisation technique that does not merely attempt to copy phrases from the original text that are deemed to be valuable but attempts to paraphrase and generate new phrases that are relevant to the source material [27].

In past years, abstractive text summarisation has mainly been applied to structured text, for example news articles, scientific publications, online reviews and standard text documents, where the flow of information is very clearly segmented in paragraphs [49]. However, dialogue or ASR output is often informal, repetitive, and complete with hesitations, interruptions, mispronunciations along with a plethora of noise. Main ideas are scattered across the conversation and repeated across different speakers, which significantly increases the difficulty of successfully summarising such transcripts [8]. Moreover, the research that has attempted multi-speaker conversation summarisation has primarily attempted to process short interactions between people, as opposed to real-world scenarios, such as meetings and interviews, in which transcripts generally comprise thousands of words [50]. This research gap had been caused by a lack of standardised datasets, which has recently been addressed by the release of *ConvoSumm* by researchers at Facebook AI [13].

Text summarisation is predominantly performed via Seq2Seq (sequence-to-sequence) learning, in which a neural network is trained to map input sequences to output sequences [36]. The combination of transformer-based neural language models with self-supervised pre-training, such as BERT [12] and GPT-2 [32] has proven to be an effective framework and the aforementioned models have achieved state-of-the-art performance on a wide range of NLP tasks, such as text classification, named entity recognition and machine translation. Recently, large self-supervised pre-trained models that have built upon these models, such as BART [22] and PEGASUS [46], have translated this success to summarisation tasks.

BART (Bidirectional and Auto-Regressive Transformer) [22] from Facebook AI, is a Seq2Seq architecture combining a bidirectional encoder (such as BERT [12]) with a left-to-right decoder (for example, GPT-2 [32]). The process of pre-training BART involves purposely corrupting (or noising) text followed by training a model to denoise and reconstruct the original text. It is most effective when fine-tuned for text generation and Seq2Seq tasks, exhibiting a significant improvement over previous state-of-the-art results when performing abstractive dialogue summarisation. Meanwhile, PEGASUS (short for Pre-training with Extracted Gap-sentences for Abstractive Summarisation Seq2Seq models) [46], proposed by Google AI, is another abstractive summarisation model that employs Gap Sentences Generation (GSG). In GSG, entire sentences are removed from a given input sequence. Subsequently, the model is trained to predict these “masked” sentences. While GSG appears to be a challenging task, such that even the authors deemed it to be near-impossible, such training elicits a deeper understanding, particularly when the sentences chosen to be masked possess a higher importance. GSG is applied simultaneously with Masked Language Modelling (MLM) in order to train the Transformer. As a result, PEGASUS has generated summaries with an accuracy similar to humans.

Both state-of-the-art models are open-sourced and widely available to use and fine-tune using the HuggingFace Transformers [43] library. It is my intention to attempt to use and pre-train both models for conversation summarisation on the *ConvoSumm* dataset and compare and evaluate their performances, selecting the appropriate fit for the proposed system.

3 Proposed Solution & Methodology

3.1 Transcript Generation

The implementation phase can be split into two parts, the first of which includes generating a full clean transcript of the meeting or conversation, segmented by speakers. This involves two main challenges as outlined above: ASR and Speaker Diarisation. In the proposed approach, diarisation will be undertaken as the first task followed by ASR. In this way, the speaker labels and time codes will be extracted for the entire recording and the latter will then be used to transcribe the audio file segment by segment. This avoids unnecessarily aiming to transcribe segments of audio containing no speech, resulting in a more efficient process and more accurate output.

The Pyannote library [4] will be used to perform speaker diarisation on the meeting files from the AMI (Augmented Multi-party Interaction) Meeting corpus [24], as this library contains a model pre-trained on this dataset. A system will be built in which users can upload their own custom audio, or choose a meeting file from the AMI corpus to transcribe and summarise.

When introducing EESEN [25], Miao et al. highlighted one of the issues with E2E ASR as the “community lack[ing] a shared experimental platform for the purpose of benchmarking”. Recently, this has been addressed by HuggingFace transformers [43]. This library will be utilised to access, fine-tune and deploy Wav2Vec 2.0 [3]. The base and large models will be compared for this particular task and assessed on efficiency and accuracy.

3.2 Summary Generation

The second task tackles the generation of a summary of the transcript obtained. This process is reliant on the quality of the ASR output, therefore it is imperative that if this is not deemed adequate, solutions such as text enhancement or spell checking are carried out. As discussed above, the two models that will be experimented with are BERT [12] and PEGASUS [46]. These will be fine-tuned on the *ConvoSumm* [13] dataset. Different approaches will be attempted including and excluding speaker segmentation and the output will be compared, judging based on conciseness and coherency.

4 Evaluation

In order to evaluate ASR output, the Word Error Rate (WER) metric is generally used [28]. It is calculated using the formula below, in which I represents the words inserted, D refers to those deleted and S substituted from the total amount of words N.

$$WER = \frac{(I + D + S)}{N}$$

As for diarisation, pyannote.audio’s inbuilt metrics module offers the functionality to calculate diarisation error rate (DER) as well as diarisation completeness, purity and coverage [4]. There are no straightforward currently-used methods to evaluate summaries that are not problematic, therefore a subjective approach will be adopted instead. A possible method is asking users to rate the summaries generated along with the transcript in order to assess the system output and evaluate whether it has performed sufficiently or otherwise.

5 Conclusion

This progress report serves as an overview of the project, describing the general problem as well as each sub-task and the intended methods to approach them. A general review of the fields of ASR, Speaker Diarisation and Text Summarisation has been carried out and the state-of-the-art for the respective fields have been identified and evaluated. This process will continue at greater length and depth in the dissertation. In the implementation phase it is envisaged that it will be discovered how these different fields can be utilised together in industry to tackle the real-world application that is note-taking during meetings, building upon existing research that addresses this. It is imperative to note that while systems

such as the one proposed in this work provide several benefits for teams, such as enhanced productivity and maximising individual performance, the recording of conversations and subsequent processing of user data by algorithmic tools may raise a variety of concerns regarding privacy, psychological safety and consent [6].

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